UK LOGISTICS

INFRASTRUCTURE DISTRIBUTION NETWORK INTERACTIVE MAP

PROJECT REPORT

CREATED & VISULIZATION BY ARTHUR ZHANG

STUDENT NUMBER: 21005241

UK LOGISTICS

INFRASTRUCTURE DISTRIBUTION NETWORK

PROJECT REPORT

CREATED & VISULIZATION BY ARTHUR ZHANG

ABOUT THIS PROJECT

This visualisation aims to explore the spatial distribution of logistics infrastructure across the United Kingdom and to reveal how logistics activity is organised at multiple spatial scales. Although often framed as neutral infrastructure, logistics facilities are unevenly distributed across the UK, mirroring—and in some cases amplifying longstanding North – South regional disparities (Hesse, 2008)[1]. By mapping facility locations nationally, the project seeks to identify broad corridors of logistical concentration, urban clusters, and areas of relative absence, rather than to evaluate service quality.

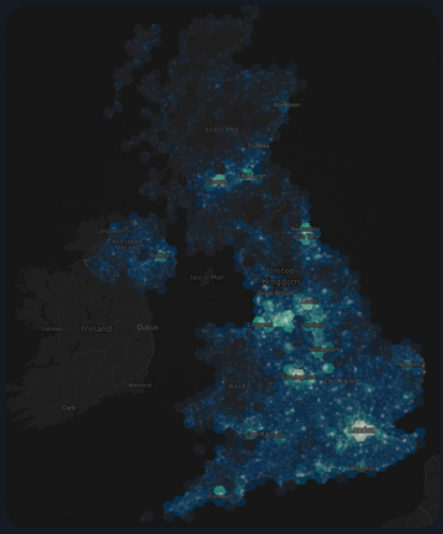
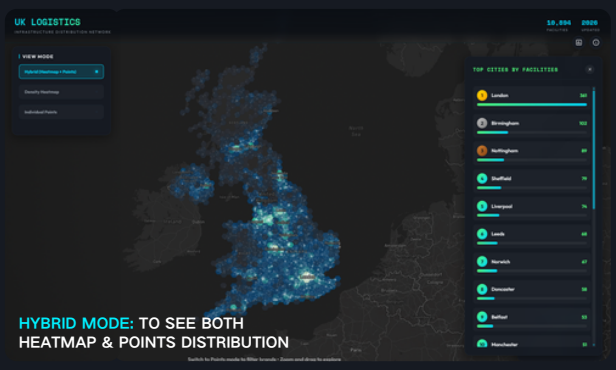


Fig 1: Main Map in this website (Heatmap + Points)

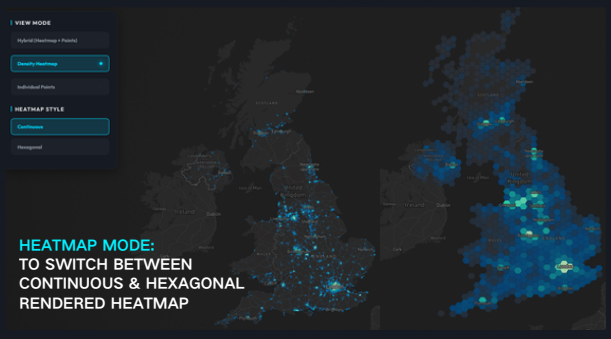
The visualisation is designed as an exploratory tool. Users can move between national and local scales, compare different density representations, and filter facilities by brand. This approach encourages spatial interpretation and pattern recognition, allowing users to consider how logistics infrastructure relates to urbanisation, transport networks, and regional structure, while acknowledging the limits of the underlying data.

FEATURES IN THIS WEBSITE

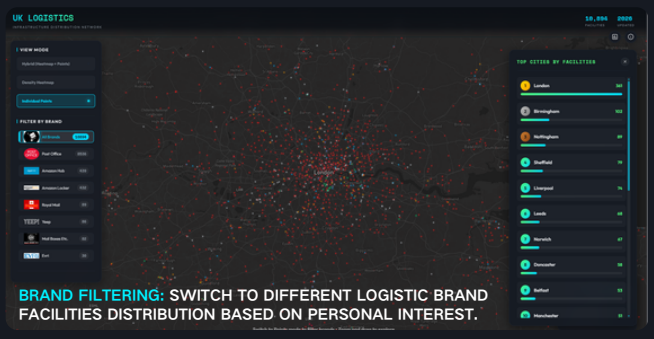


HYBRID MODE: TO SEE BOTH HEATMAP & POINTS DISTRIBUTION

FLY FEATURE: SWITCH TO CITY CENTER AFTER CLICKING CITY NAME AT THE CHARTS SECTION



HEATMAP MODE: TO SWITCH BETWEEN CONTINUOUS & HEXAGONAL RENDERED HEATMAP



BRAND FILTERING: SWITCH TO DIFFERENT LOGISTIC BRAND FACILITIES DISTRIBUTION BASED ON PERSONAL INTEREST.

Fig 2: Main Interactive Feature in this Website

DESIGN & TECHNICAL APPROACH

The map interface is built as a web-based interactive visualisation using Mapbox GL JS. Two complementary approaches to representing spatial density are implemented. First, a continuous density (heatmap-style) view provides an intuitive overview of regional intensity, smoothing individual locations to highlight large-scale patterns and corridors. Additionally, a hexagonal density view aggregates facilities into equal-area hexagonal cells, enabling clearer comparison between regions and reducing visual bias caused by uneven point concentrations in major cities (Battersby et al., 2017)[2].

The hexagonal approach was chosen to address limitations observed in continuous density visualisations, particularly the tendency for dense urban areas to dominate visual attention while masking sparsely populated regions. By aggregating facilities into fixed spatial units, the hexagonal view makes areas of concentration and absence more explicit and comparable. Individual facility points are revealed only at higher zoom levels, ensuring a clear visual hierarchy in which aggregated density communicates structure and points communicate presence.

The interface also supports interactive filtering by logistics brand, allowing users to switch between subsets of facilities. This functionality enables comparative exploration of how different operators distribute their infrastructure spatially, without implying completeness or competitive dominance. All visual design choices prioritise legibility, consistency across zoom levels, and transparency in what each representation is intended to show.

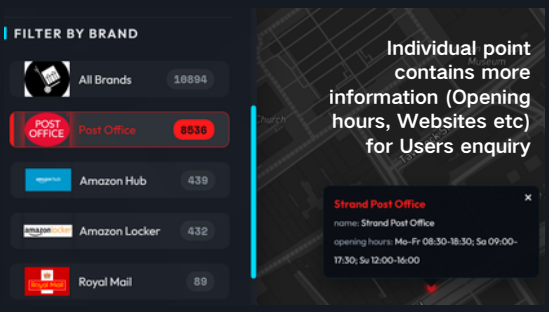


Fig 3: Further Interactive Feature

DATA SOURCES & LIMITATIONS

Facility location data are derived from OpenStreetMap, a crowd-sourced global geographic database maintained by volunteer contributors. The dataset was extracted from Overpass Turbo[3], cleaned, and processed to identify logistics-related facilities across the UK. Brand classifications were derived through attribute parsing and manual standardisation and should be interpreted as indicative rather than authoritative.

As a open-sourced dataset, OpenStreetMap exhibits spatial and thematic variation in completeness, particularly between urban and rural areas. The visualisation therefore does not claim to represent a complete inventory of logistics infrastructure, or does density imply capacity, throughput, or service quality. Instead, the map reveals a spatial patterns of logistical concentration, connectivity, while highlighting both the analytical potential and limitations of open geographic data.

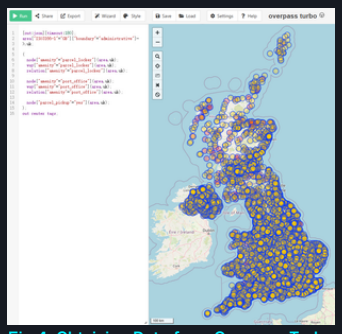


Fig 4: Obtaining Data from Overpass Turbo

[1]: Hesse, M. (2008) The City as a Terminal: The Urban Context of Logistics and Freight Transport. *Aldershot*: Ashgate.

[2]: Battersby, S.E., Jenny, B., Weber, A. and McGill, S. (2017) 'An evaluation of grid-based visualizations for geographic data', *Cartography and Geographic Information Science*, 44(5), pp. 393–414.

[3]: Overpass Turbo (2025) Overpass Turbo: A web-based data mining tool for OpenStreetMap. Available at: *https://overpass-turbo.eu*